

Docker: Build, Ship and Run Any App, Anywhere

This article introduces the reader to various aspects of Docker, starting with how to install it.



Docker is basically a container management service. Some of the most common and important terms associated with it are containers, image, package, deploy, ship, etc. Docker is very handy for developing applications and software, and to then ship them into containers that can be deployed anywhere, irrespective of the platform. Importantly, Docker is open source, which means that anyone in the world can contribute to it and manipulate the code to meet one's own needs if additional features that aren't available are required.

Why Docker?

- Docker has the ability to reduce the size of the deployment by providing a smaller footprint on the operating system via containers.

- With containers, it becomes easier for groups of people across different units, such as the development team, to work seamlessly across applications.
- Docker containers can be deployed with your applications, anywhere, on any physical and virtual machine.

Installing Docker in Linux

The following steps can be followed to install Docker in Ubuntu.

Step 1: Docker is only designed to work on Linux kernel version 3.8 and later, for which you need to check the kernel version. That can be done by using the following command:

```
$ uname -a
```

If your Linux kernel version is earlier than 3.8, then upgrade it using

the following command:

```
$ sudo apt-get update
```

Step 2: The next step is to install the necessary certificates that will be required to work with the Docker site later on, in order to download the necessary Docker packages. This can be done with the following command:

```
$ sudo apt-get install apt-transport-https ca-certificates
```

Step 3: Next, add the new GPG key, which is required to ensure that all data is encrypted when downloading the necessary packages for Docker.

The following command will download the key with the ID `58118E89F3A912897C070ADBF76221572C52609D` from the key server `hkp://ha.pool.sks-keyservers.net:80` and add it to the `adv` keychain. Do note that this particular key is required to download the necessary Docker packages.

```
$ sudo apt-key adv \ --keyserver hkp://ha.pool.sks-keyservers.net:80 \ --recv-keys 58118E89F3A912897C070ADBF76221572C52609D
```

Step 4: Next, depending on the version of Ubuntu you have on your device, you will need to add the relevant site to `docker.list` for the package manager to detect the Docker packages from the Docker site, and download them accordingly. The